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Oefeningenreeks 3A nr. 37.11

$$\begin{aligned} f(x) &= \ln \sqrt{\frac{1 + \cos(ax)}{1 - \cos(ax)}} \\ &= \frac{1}{2} \ln \left(\frac{1 + \cos(ax)}{1 - \cos(ax)} \right) \\ &= \frac{1}{2} [\ln(1 + \cos(ax)) - \ln(1 - \cos(ax))] \\ f'(x) &= \frac{1}{2} \left[\frac{-a \sin(ax)}{1 + \cos(ax)} - \frac{a \sin(ax)}{1 - \cos(ax)} \right] \\ &= \frac{-a \sin(ax)}{2} \left[\frac{1}{1 + \cos(ax)} + \frac{1}{1 - \cos(ax)} \right] \\ &= \frac{-a \sin(ax)}{2} \left[\frac{2}{1 - \cos^2(ax)} \right] \\ &= \frac{-a \sin(ax)}{2} \cdot \frac{2}{\sin^2(ax)} = -a \sin(ax) \cdot \frac{1}{\sin^2(ax)} = -\frac{a}{\sin(ax)} \end{aligned}$$

References